Chapter 35 Review

Campbell Biology, AP Edition, 12e

Chapter 35 Review

Summary of Key Concepts

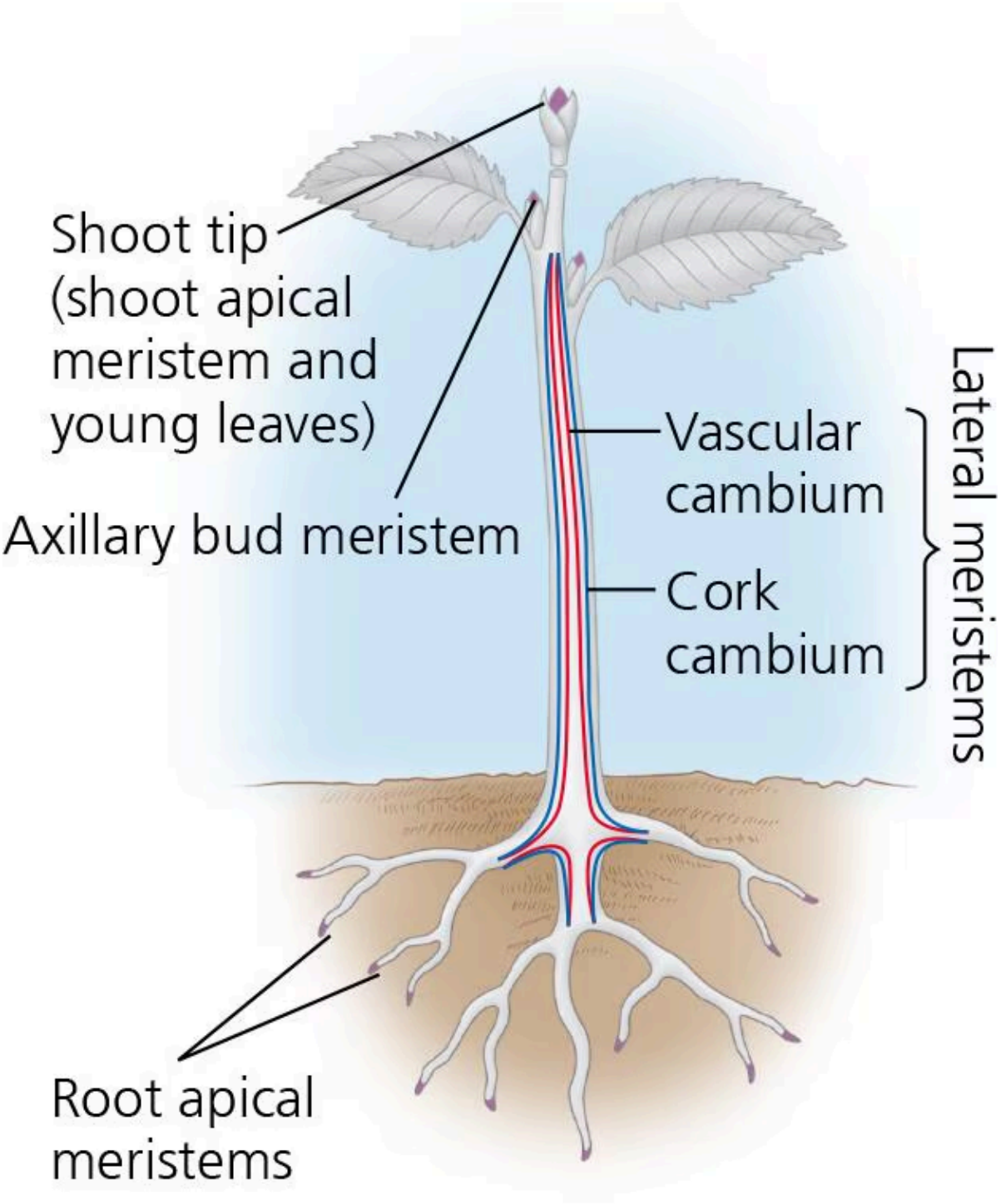
Studies with *Arabidopsis* confirm that asymmetrical cell division leads to the formation of stomata and that the regulators of stomata formation are themselves regulated by environmental changes. How would environmental interactions control the formation of stomata? **(BIG IDEA 4)**

Concept 35.1: Plants have a hierarchical organization consisting of organs, tissues, and cells

- Vascular plants have shoots consisting of **stems**, **leaves**, and, in angiosperms, flowers. ****Roots** anchor the plant, absorb and conduct water and minerals, and store food. Leaves are attached to stem **nodes** and are the main **organs** of photosynthesis. The **axillary buds**, in axils of leaves and stems, give rise to branches. Plant organs may be adapted for specialized functions.
- Vascular plants have three **tissue systems**—dermal, vascular, and ground—which are continuous throughout the plant. The **dermal tissue** is a continuous layer of cells that covers the plant exterior. **Vascular tissues** (**xylem** and **phloem**) facilitate the long-distance transport of substances. **Ground tissues** function in storage, metabolism, and regeneration.
- **Parenchyma cells** are relatively undifferentiated and thin-walled cells that retain the ability to divide; they perform most of the metabolic functions of synthesis and storage. **Collenchyma cells** have unevenly thickened walls; they support young, growing parts of the plant. **Sclerenchyma cells**—sclereids and fibers—have thick, lignified walls that help support mature, nongrowing parts of the plant. ****Tracheids** and **vessel elements**, the water-conducting cells of xylem, have thick walls and are dead at functional maturity. **Sieve-tube elements** are living but highly modified cells that are largely devoid of internal organelles; they function in the transport of sugars through the phloem of angiosperms.

Describe at least three specializations in plant organs and plant cells that are adaptations to life on land.

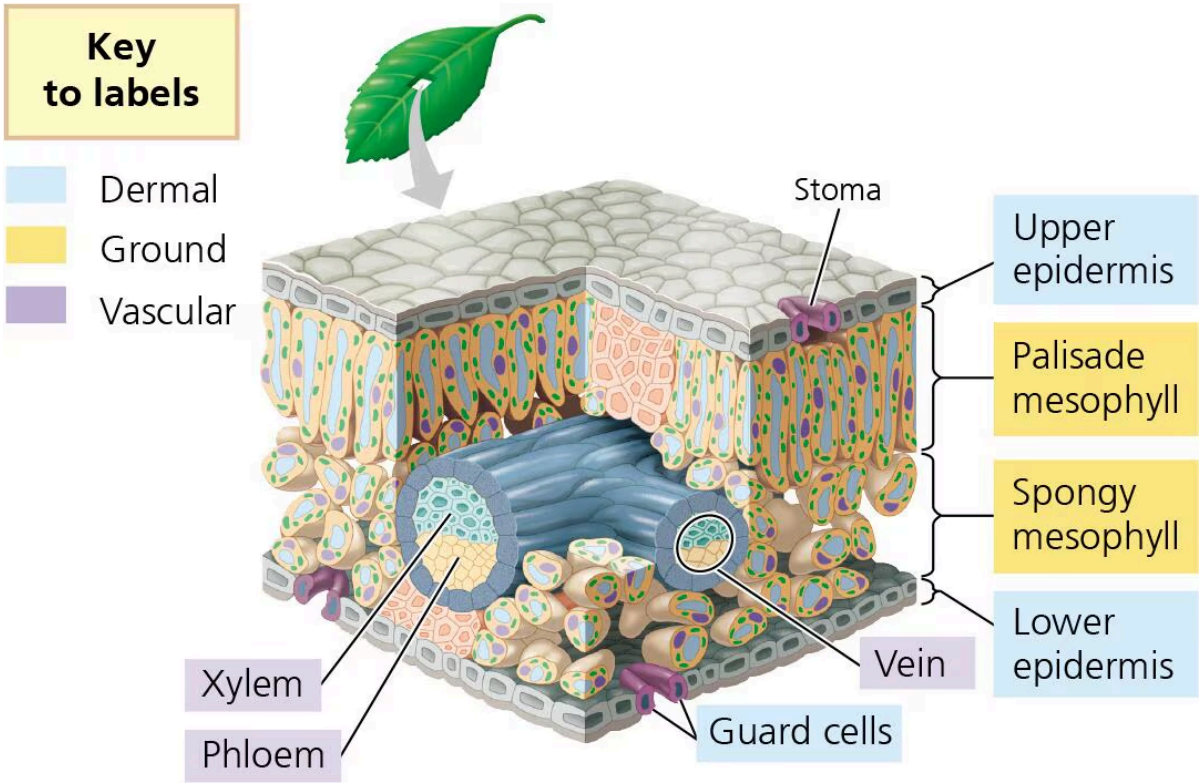
Concept 35.2: Different meristems generate new cells for primary and secondary growth



What is the difference between primary and secondary growth?

Concept 35.3: Primary growth lengthens roots and shoots

- The root **apical meristem** is located near the tip of the root, where it generates cells for the growing root axis and the **root cap**.
- The apical meristem of a shoot is located in the **apical bud**, where it gives rise to alternating **internodes** and leaf-bearing nodes.
- Eudicot stems have vascular bundles in a ring, whereas monocot stems have scattered vascular bundles.
- **Mesophyll** cells are adapted for photosynthesis. **Stomata**, epidermal pores formed by pairs of **guard cells**, allow for gaseous exchange and are major avenues for water loss.

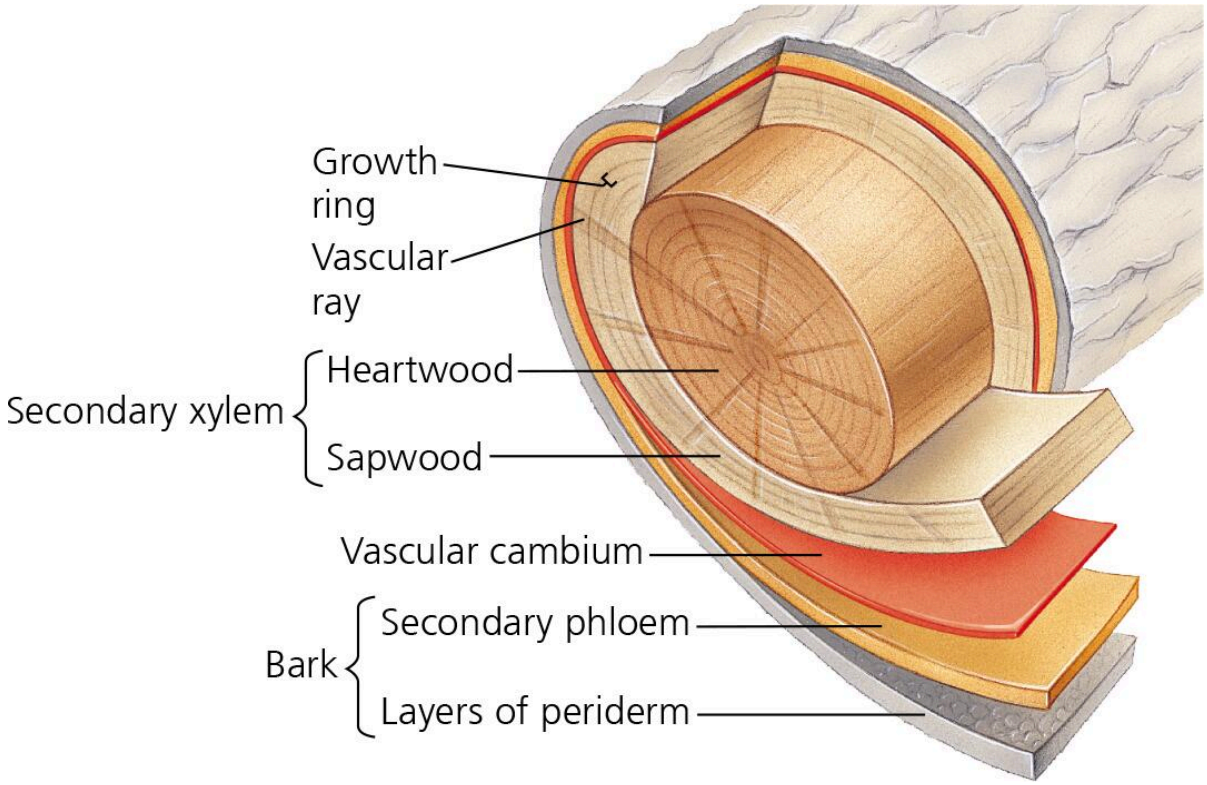


A cutaway diagram of leaf tissues shows vein at the center of the leaf, with the bundle sheath cells around the xylem and phloem tissues inside the vein. Outside the vein are the Sclerenchyma fibers. On the surface of the leaf is the waxy cuticle dotted by stoma (openings) flanked by guard cells. The upper epidermis is the top outer leaf layer, followed by the palisade mesophyll inside and then the spongy mesophyll. The lower epidermis is at the bottom of the leaf.

How does branching in roots differ from that in stems?

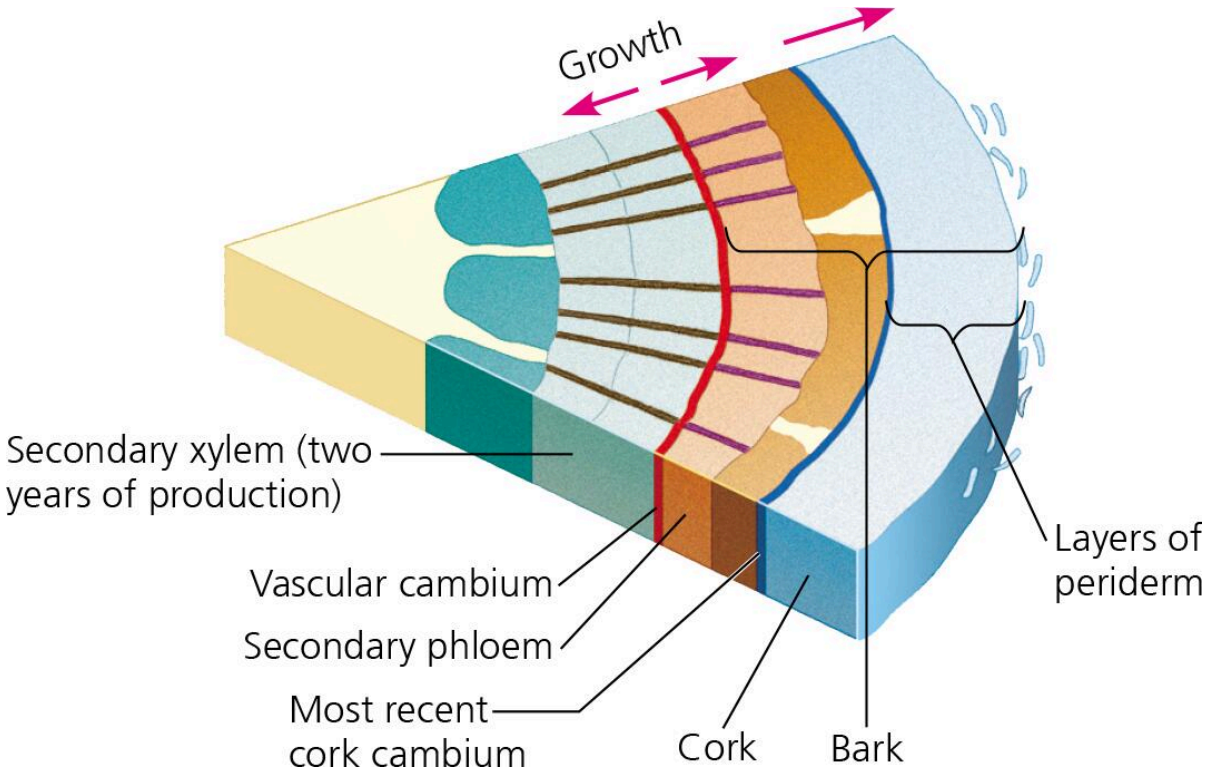
Concept 35.4: Secondary growth increases the diameter of stems and roots in woody plants

- The **vascular cambium** is a meristematic cylinder that produces secondary xylem and secondary phloem during ****__secondary growth__. Older layers of secondary xylem (heartwood) become inactive, whereas younger layers (sapwood) still conduct water.



Cross-section of tree trunk showing the secondary xylem, including the heartwood (center of tree) surrounded by the sapwood. The vascular cambium is on the outside of the sapwood followed by two layers of bark: secondary phloem (inside) and layers of periderm (outside). Ring growth is highlighted by a ring in the sapwood and vascular rays are shown radiating out from the center of the tree.

- The **cork cambium** gives rise to a thick protective covering called the periderm, which consists of the cork cambium plus the layers of cork cells it produces.



The cross-section has a layer of epidermis on the outside followed by the cork. The next ring is the most recent cork cambium followed by the secondary phloem and vascular cambium. The next ring is of the secondary xylem (two years of production) followed by a ring of the innermost layer. Growth, shown as a double-headed arrow, occurs outward from the secondary phloem onward.

What advantages did plants gain from the evolution of secondary growth?

Concept 35.5: Growth, morphogenesis, and cell differentiation produce the plant body

- Cell division and cell expansion are the primary determinants of growth.
- Morphogenesis, the development of body shape and organization, depends on cells responding to positional information from their neighbors.
- Cell differentiation, arising from differential gene activation, enables cells within the plant to assume different functions despite having identical genomes. The way in which a plant cell differentiates is determined largely by the cell's position in the developing plant.
- Internal or environmental cues may cause a plant to switch from one developmental stage to another—for example, from developing juvenile leaves to developing mature leaves. Such changes in shape are called **phase changes**.
- Studies of floral development have provided a model system for studying ***pattern formation***. ***The ABC*****hypothesis** identifies how three classes of genes control formation of sepals, petals,

stamens, and carpels.

By what mechanism do plant cells tend to elongate along one axis instead of expanding in all directions?